



Open19 System Specification

Linux Foundation Sustainable Scalable Infrastructure Alliance

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Alliance

The Sustainable & Scalable Infrastructure Alliance (SSIA) is a community-based organization dedicated to creating, fostering and driving adoption of open standards that increase the efficiencies of the data center and digital infrastructure supply chains. We enable digital leaders from any sized organization to adopt and deploy advanced and highly efficient rack-level datacenter technologies to increase sustainability, power next-generation workloads and reduce cost and waste, regardless of deployment size or location.

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Chapter 1. Change log

Author	Date	Version	Change Details
My Truong	Nov 2023	2.0	Public Spec release

Definitions

ASHRAE

American Society of Heating, Refrigerating and Air-Conditioning Engineers

ASHRAE W27

ASHRAE specification for water temperature range from 2 °C to 27 °C

OCP

Open Compute Project

SFP

SNIA SFF-8402 SFP+ 1X Pluggable Transceiver Solution

SNIA

Storage Networking Industry Association

QSFP

SNIA SFF-8679 QSFP+ 4X Pluggable Transceiver Solution

QSFP-DD

QSFP-DD MSA QSFP DOUBLE DENSITY 8X AND QSFP 4X PLUGGABLE TRANSCEIVERS

Chapter 2. Open19 V2 Cages

Open19 V1 defined a base building element, the brick cage, for building the Open19 infrastructure. The cage is a completely passive element able to receive any Open19 V2 compliant device. The brick definition governs cage dimensions (and visa versa). The cages are re-configurable to accommodate any Open19 V2 compliant brick.

Open19 V2 introduces cut-outs and attachment features for liquid cooling manifolds, expanded cut-out for the larger blind-mate data connector, and expanded cut-out for the new power connector.

2.1. Cage modularity

Cages are designed to support all defined bricks. The current definition of bricks use a 2RU maximum format, enabling the following brick form-factors in each 2RU section:

- (4) Single high Brick - Half-width 1RU [SHHW]
- (2) Double high bricks - Half-width 2RU [DHHW]
- (2) Single high bricks - double-width 1RU [SHDW]
- (1) Double high brick - double-width 2RU [DHDW]

The design package supports two cage form-factors, 8RU and 4RU. Additional cage groupings can be synthetically formed from these units, including the 12RU from the V1 spec. This enables extremely small deployments of 6RU (4RU cage + 2RU power shelf) with no upper limit.

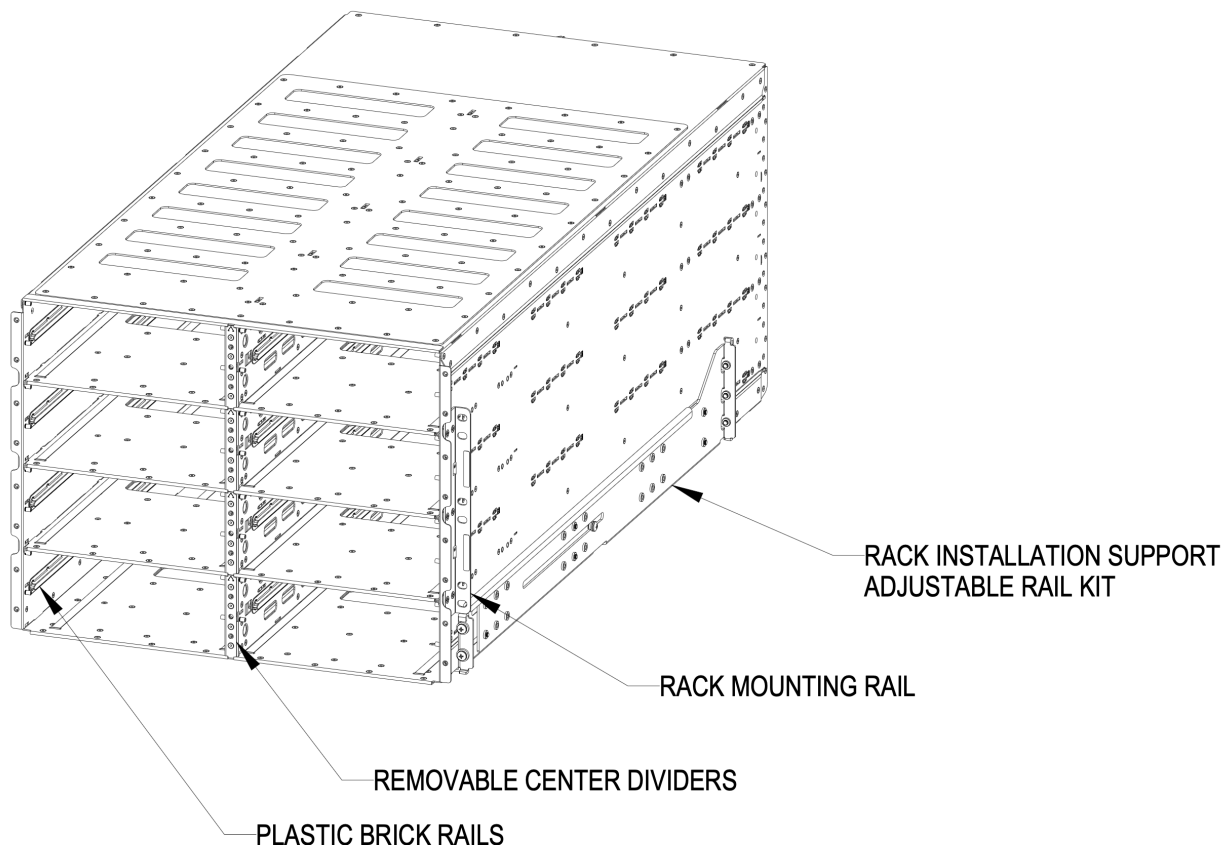


Figure 1. 8RU Open19 V2 cage front

2.2. 8RU Cage detail

The 8RU cage is designed to be installed in a standard EIA/TIA 19-inch rack. The dimensions of cage allow for proper mating of all blind-mate interfaces on bricks.

2.2.1. 8RU Cage dimensions

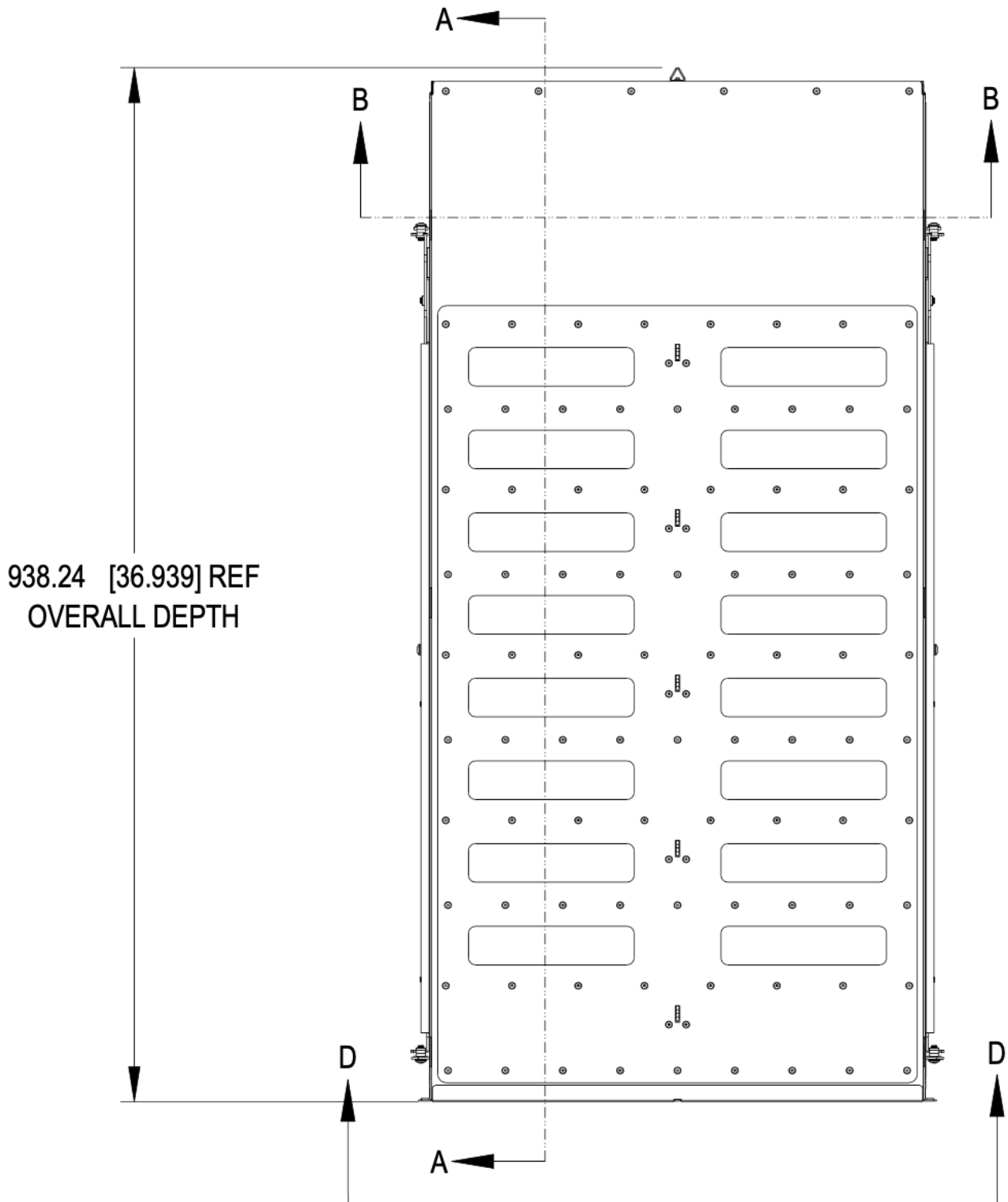


Figure 2. 8RU Open19 V2 cage depth

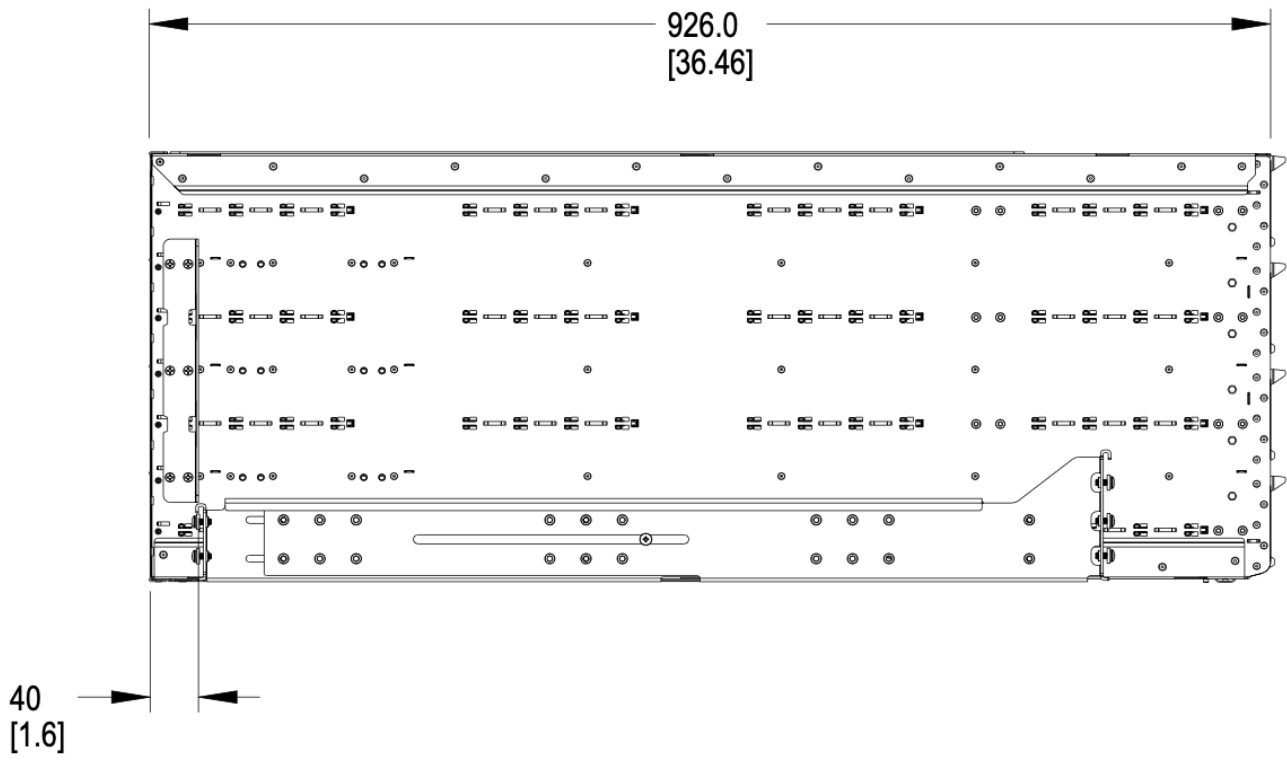


Figure 3. 8RU Open19 V2 cage depth side view

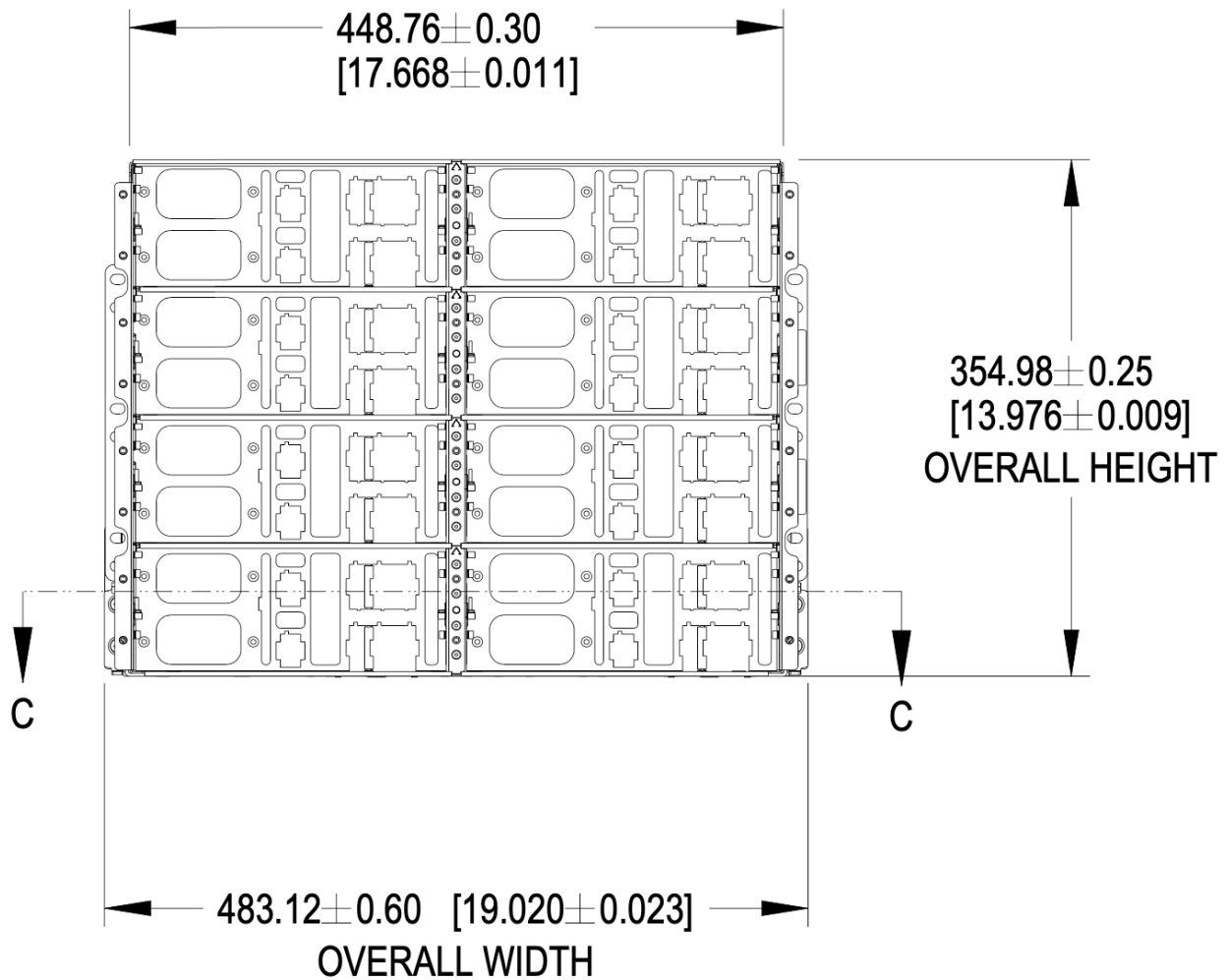


Figure 4. 8RU Open19 V2 cage rear

2.2.2. 8RU Cage partition

The 8RU cage is built around a 2RU section, 84.7 ± 0.25 mm tall and 443.80 ± 0.25 mm wide. This section can accommodate a single DHDW brick or two SHDW bricks.

2.2.3. 8RU Cage Center divider (CD) partition

With a center divider (CD) installed, each section can accommodate one of the following combinations:

- One to four SHHW bricks
- One or two DHHW bricks

Dropped from the V2 spec is one DHHW plus one or two SHHW bricks.

With the center divider installed, each 2RU section is 84.7 ± 0.25 mm tall and 215.02 ± 0.25 mm.

2.2.4. 8RU Cage Center Divider (CD)

The Center Divider (CD) is installed to support SHHW and DHHW bricks. Mechanical retention of

the divider should be integrated into the divider. The Open19 V1 specification provides details on a reference CD design.

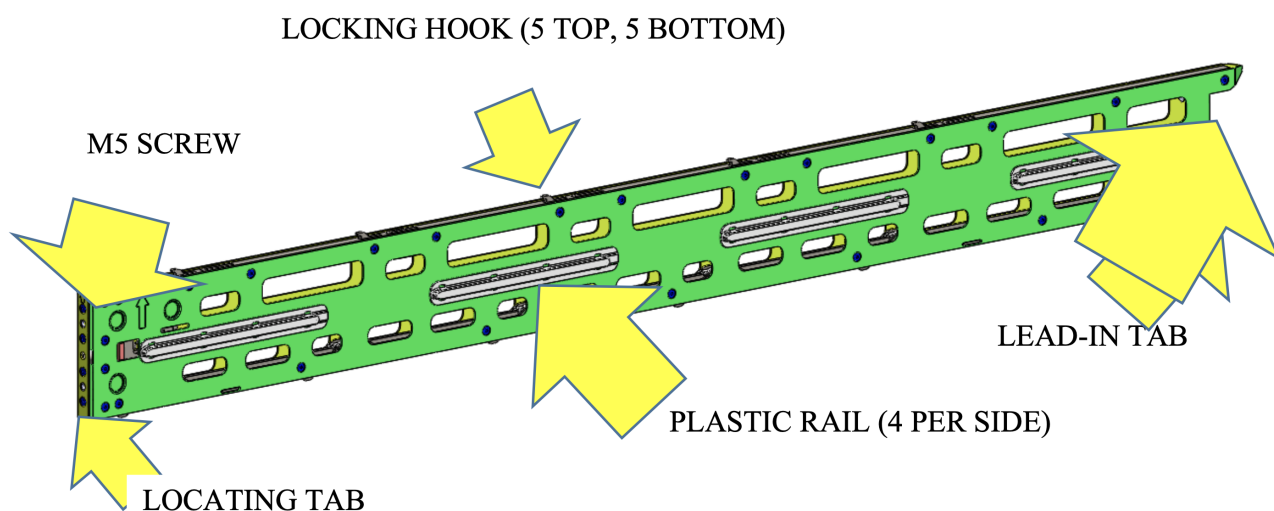


Figure 5. Open19 V2 cage Center Divider

2.2.5. Horizontal divider

The horizontal divider partitions the cage in 2RU sections. Horizontal dividers shall have strips to provide a low-friction surface and minimize metal-on-metal contact.

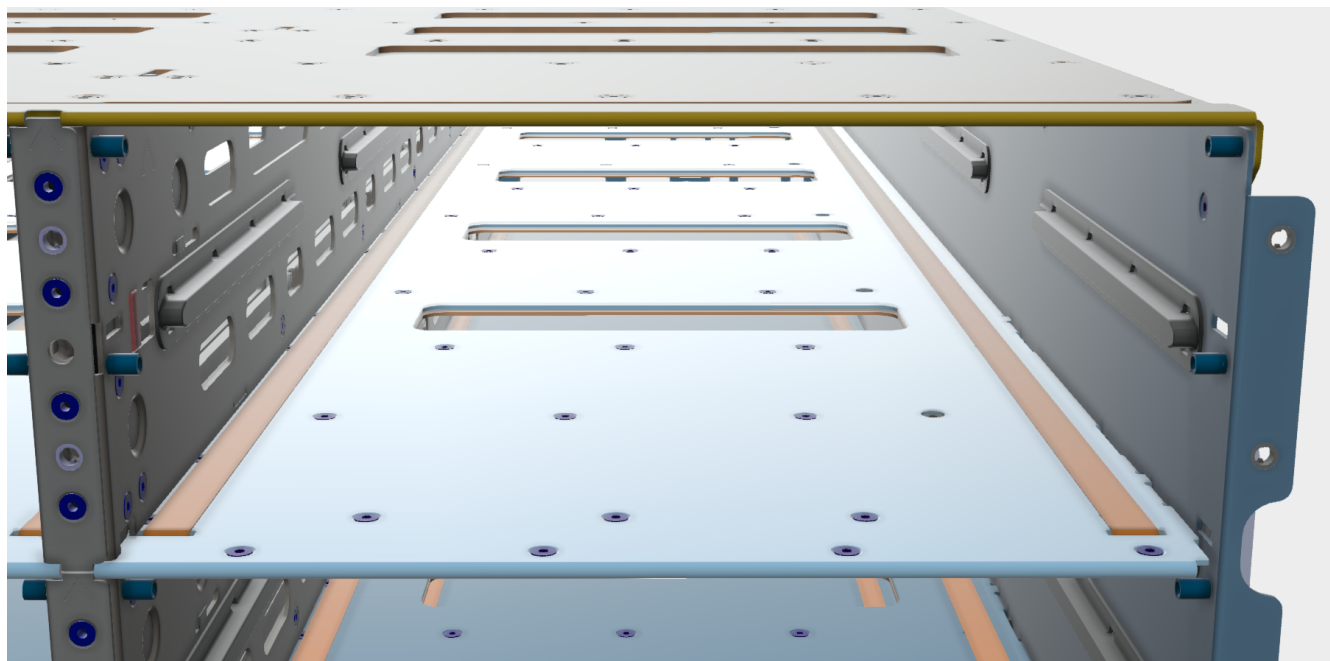


Figure 6. Open19 Cage horizontal divider

2.2.6. Load capacity

The 8RU cage can accommodate 16 bricks with a maximum 30lbs (13.6kg) each.

2.2.7. Brick Retention

Each side wall contains eight (8) Southco D9-63-11 self-clinching receptacles that secure bricks with quarter-turn fasteners as the primary means of retention.

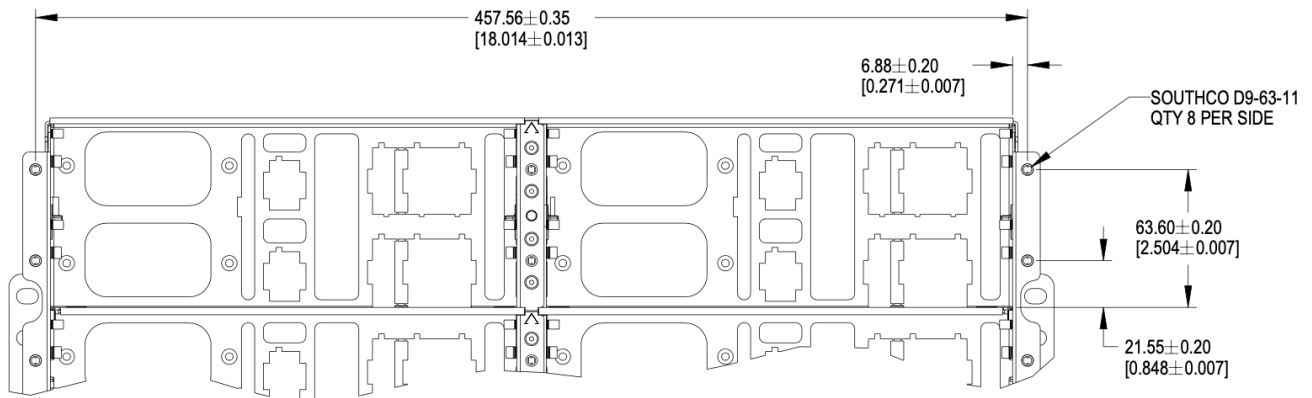


Figure 7. Cage retention quarter turn fastener locations

For liquid cooling applications, additional features are available to provide sufficient mechanical assistance to overcome total connector forces. These features are detailed in the liquid cooling section.

2.2.8. Blind mate cutouts

The rear wall of the 2RU space provides consistent cutouts for the cooling, data, and +48V DC power cables.

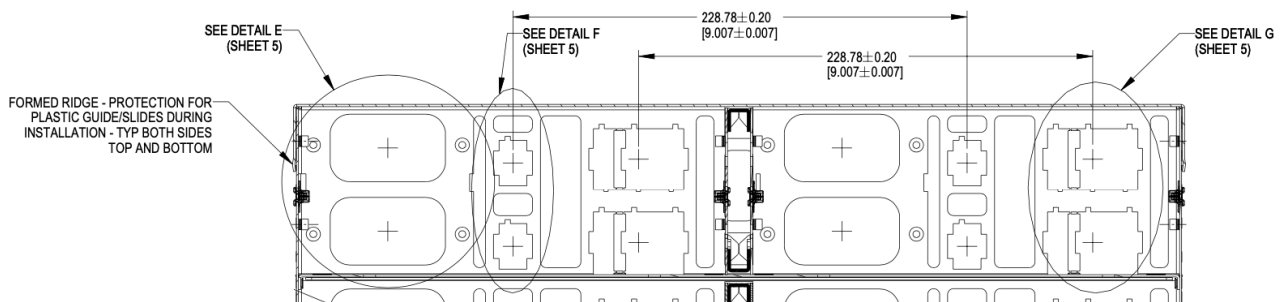


Figure 8. Cage rear panel cutouts

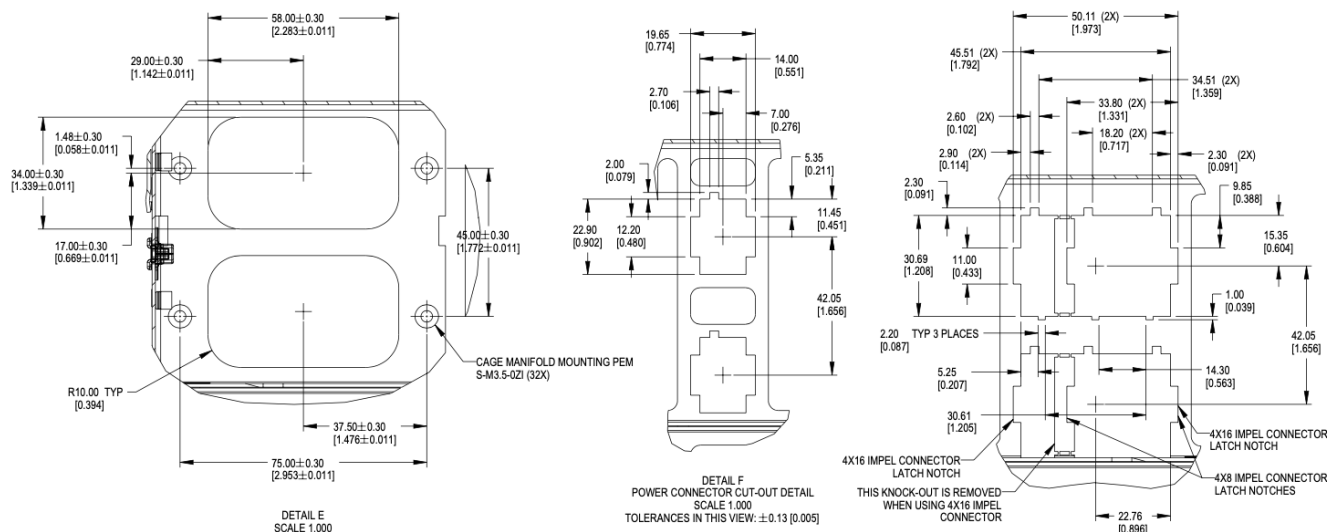


Figure 9. Cage rear panel cutout details

2.2.9. Keying stops

Keying stops are provided in the right and left rear corners of each cage section. These stops (bent tabs) interface with notches in the brick sides to provide a positive stop during brick insertion and prevent over insertion or possible connector damage. They are positioned as shown.

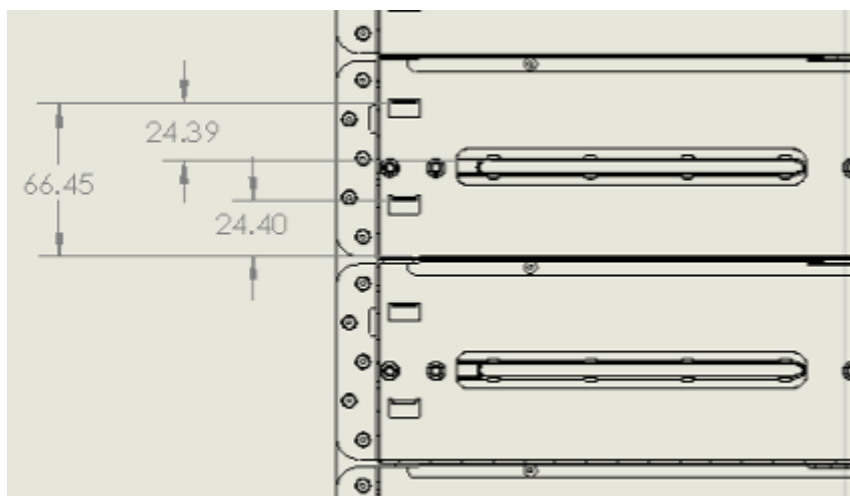


Figure 10. Brick keying stops

2.2.10. Mounting ear locations

The primary cage mounting ear is positioned 40mm behind the front face of the cage side wall. Two additional positions are provided, 100mm and 200mm from the mounting ear surface to the front face of the cage side wall.

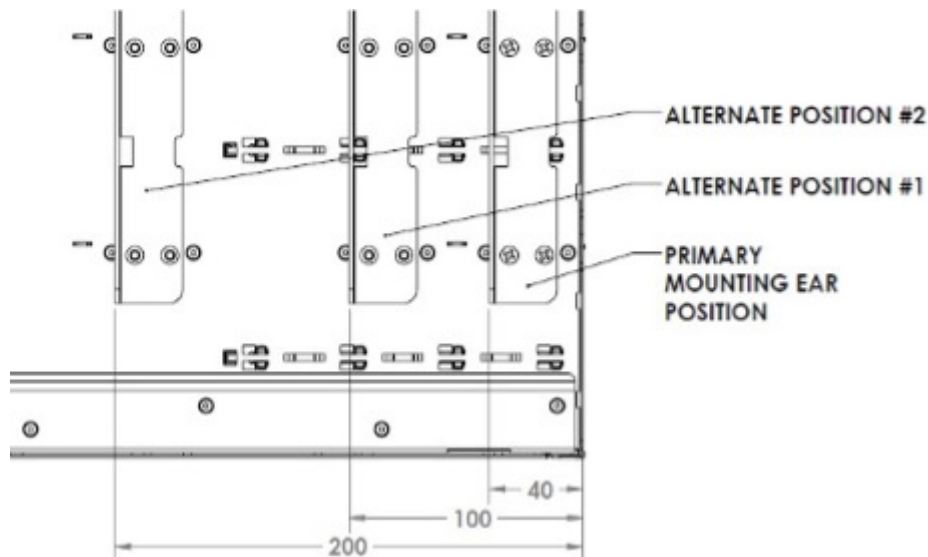


Figure 11. Cage mounting ear locations

These recommended locations are design guidance from V1 and may be adapted for specific applications.

2.2.11. Cage mounting rails

Rack rails are the primary mechanism to support the cage during installation and use. A hook facilitates rail installation in square hole four-post racks. Rails should not protrude into the bottom surface of the cage.

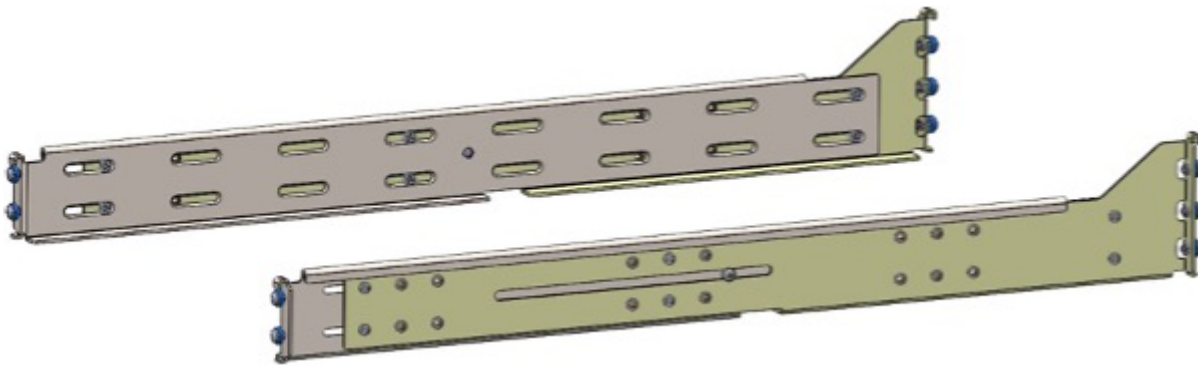


Figure 12. Cage rack rails

Installing rails should accommodate a range of rack post spacing from 28.5" to 36.0". For other types of racks, a modified version of the rail can be developed.

The rails should support:

- Bare cage installed in a rack during shipping – cage mounting ear screws must secure cage in the rack
- Bare cage installed in a data center rack – cage mounting ear screws not installed
- Fully populated cage during installation into a rack

- Fully populated cage during use – cage mounting ear screws must secure cage in the rack

2.2.12. Material

The 8RU cage is designed to be constructed of SGCC sheet metal. 1.2mm (19 gauge steel) is the primary thickness used throughout, with 0.8mm and 1.6mm used on some stiffener panels, 1.88mm used for the cage mounting brackets and 2.38mm for the rails.

The data and power connectors are built with a 1.20mm +/- 0.05mm expectation for latching mechanism. Panel thickness may be altered as required; the area surrounding the connectors shall remain 1.20mm +/- 0.05mm and shall maintain the 6.23mm face-to-face panel spacing.

Two molded plastic parts are used on the cage assembly. These parts are identical to the V1 design.

Sliders are used on the cage walls and center dividers to support bricks. Two sliders are used per brick side, 64 sliders are used in a cage fully populated with center dividers. Sliders are molded in Ultem 1000 resin.

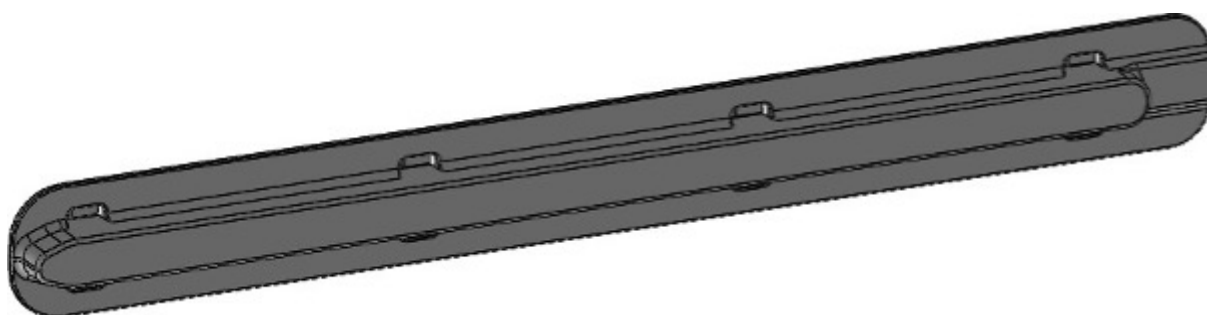


Figure 13. Cage slide

Dowel pin inserts are used in ten (10) places in each center divider. These parts are molded in Ultem 1000 resin.

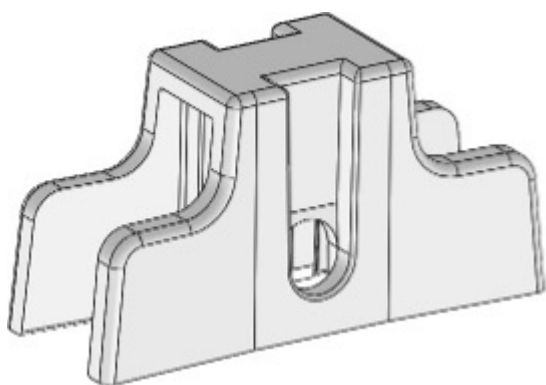


Figure 14. Cage divider

Low-friction strips used on the horizontal dividers are made of black FR700-701 polycarbonate sheet with 3M 467 PSA applied.

Chapter 3. Liquid cooling support

3.1. Brick insertion and retention assistance for liquid cooling

For liquid cooling applications, two different mechanical assist for insertion and retention are available in the cage. Bricks can implement features to make use either/both features.

The features are:

- Brick retention standoff, available on both sides in the mechanical guide channel
- Slots, available on both sides of the cage wall

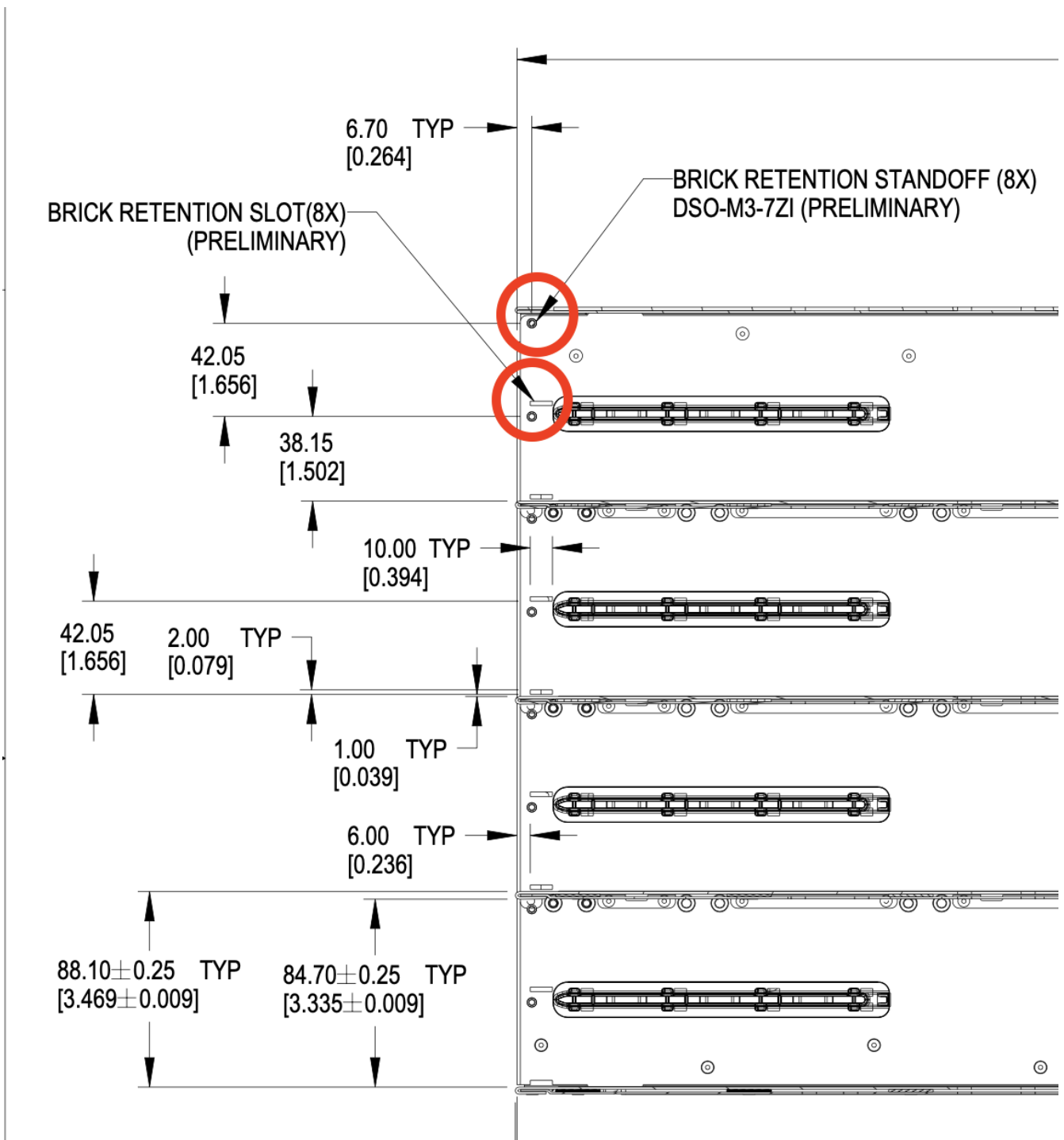


Figure 15. Brick mechanical assistance features in cage

Each feature shall support 100 lbf (445N) force or greater, using the recommended 1.3mm SGCC material.

3.2. Manifold attachments

The 8RU cage provides consistent attachment points for both single-phase and two-phase liquid cooling manifolds. PEM nuts are available in every brick location.

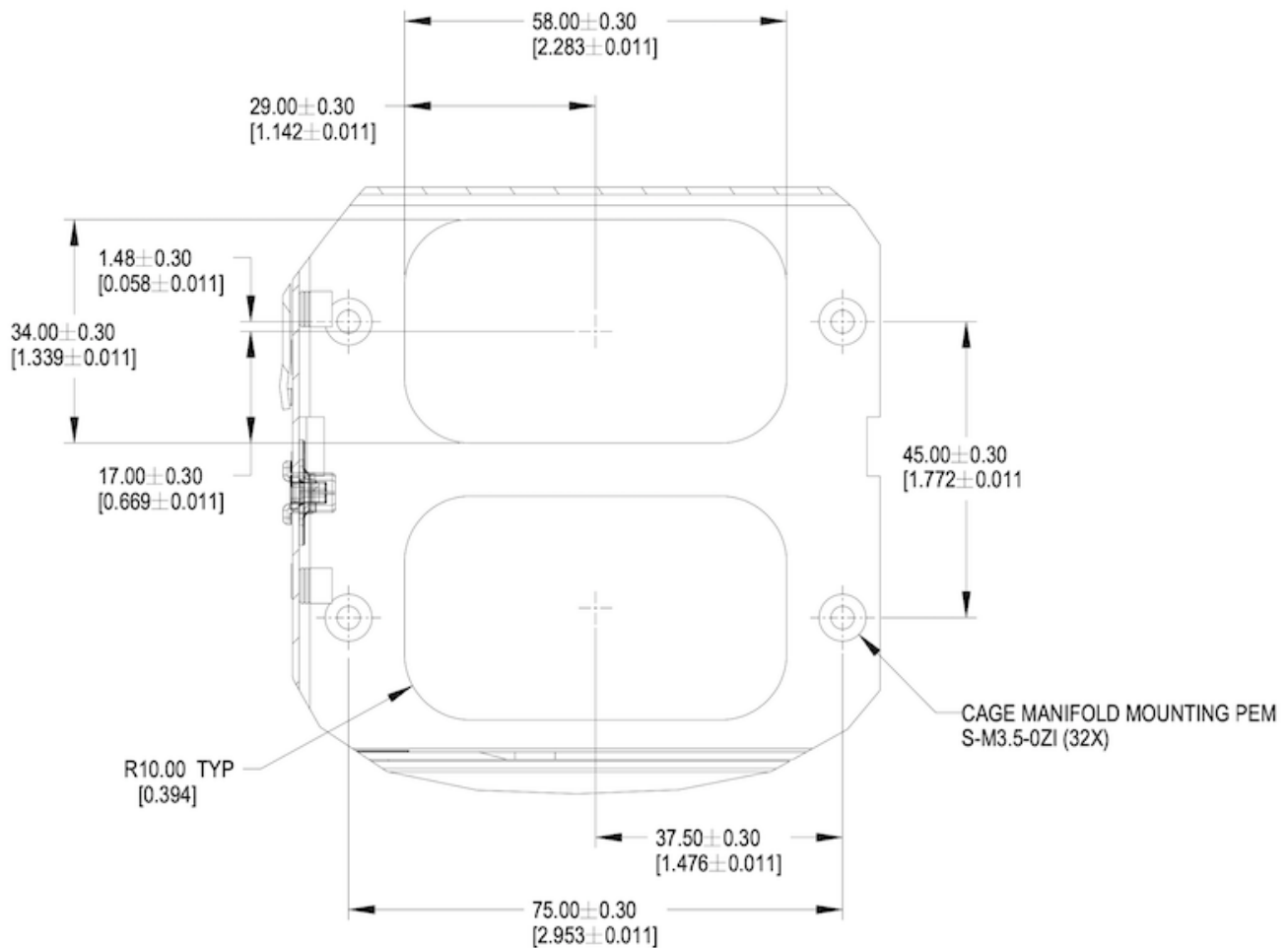


Figure 16. Cage manifold attachment

3.3. 8RU Cage Manifold

3.3.1. Attachment points

The 8RU cage provides consistent attachment points for both single-phase and two-phase liquid cooling manifolds. Two PEM nuts are available in every brick location.

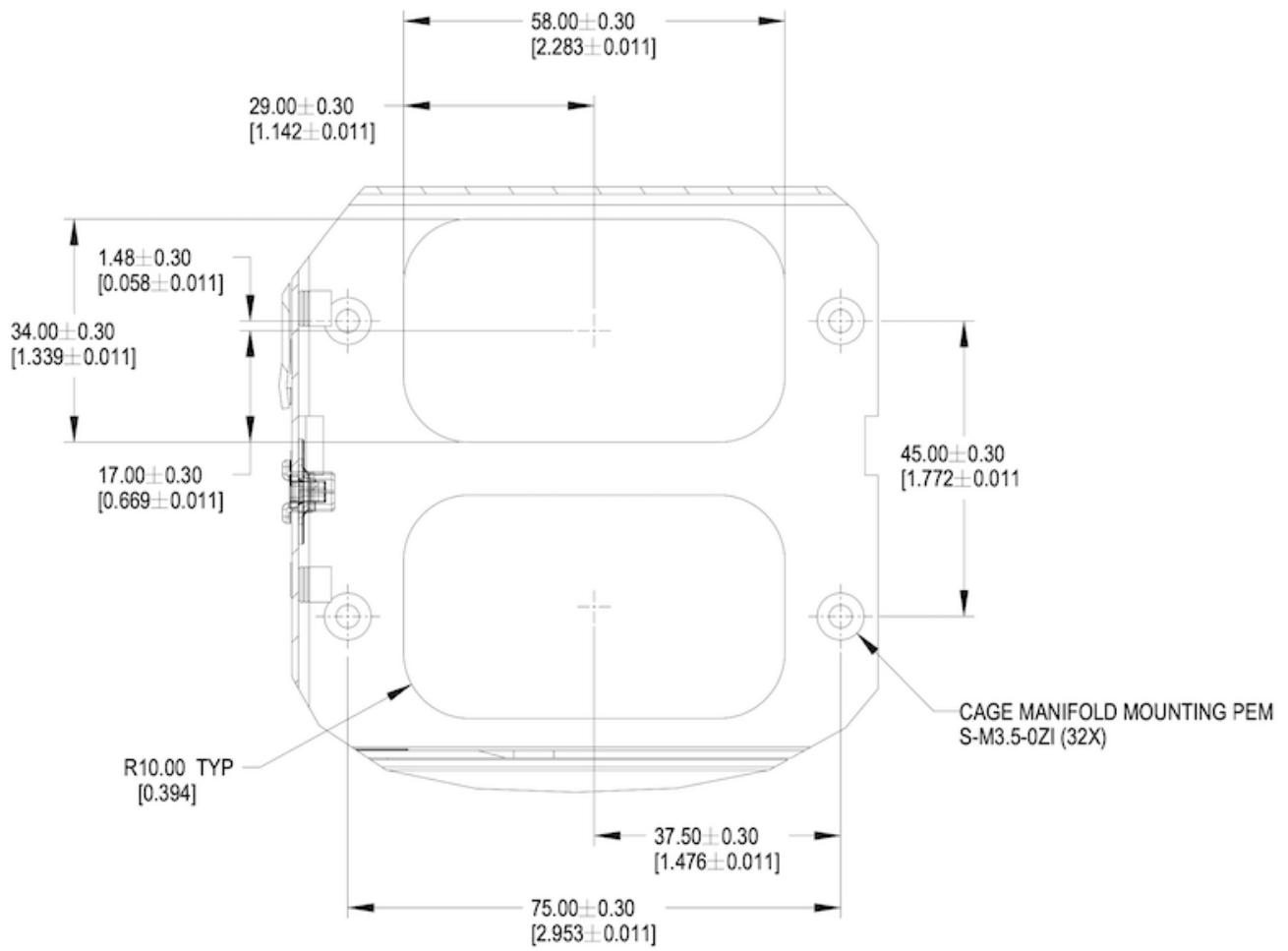


Figure 17. Manifold attachment



Figure 18. Sample left manifold bracket

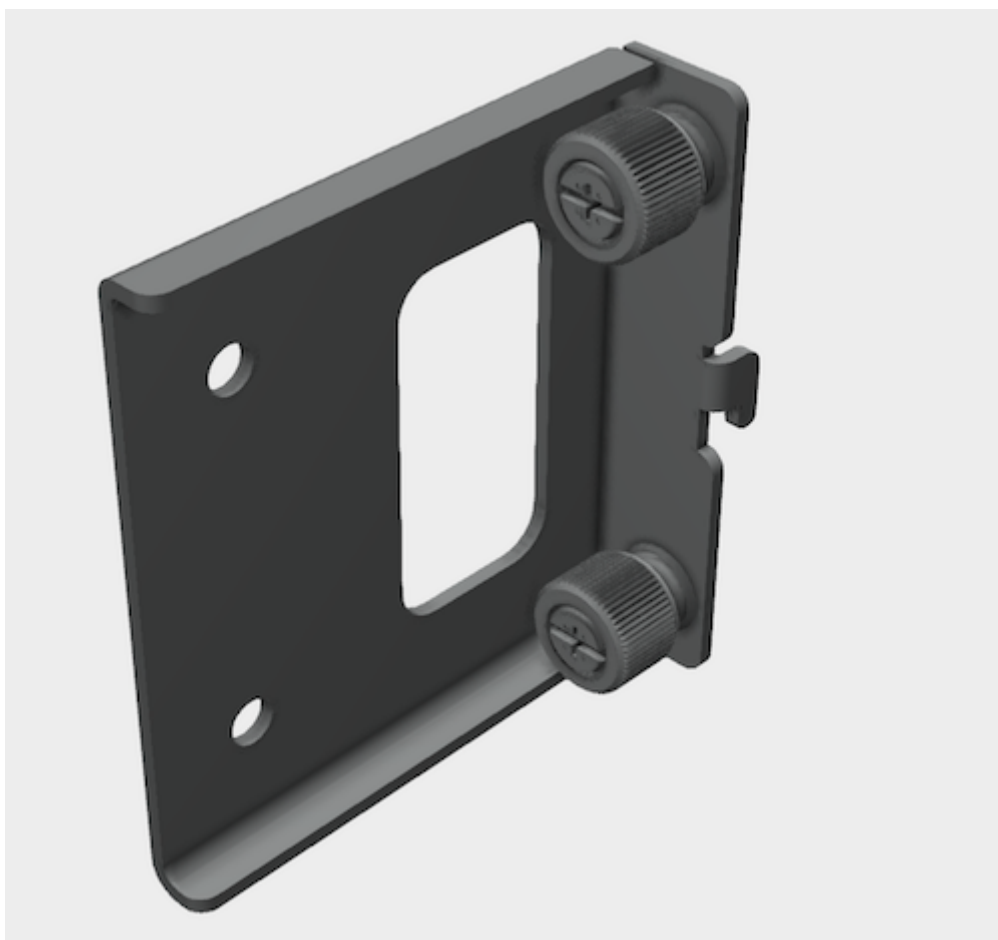


Figure 19. Sample right manifold bracket

3.3.2. Manifold

The liquid cooling manifold is not interoperable between single-phase and two-phase, by design. The bricks provide a blind-mate projection from the brick into the liquid cooling manifold.

3.3.3. Single-phase manifold

Single-phase manifold specified dimensions

The actual design and size of the manifold is up to the implementor. There are a few critical dimensions that shall be kept consistent to provide compatibility for all single-phase implementations.

Single-phase manifold connector spacing

Cage manifolds shall have female blind-mate connectors (bodies), BLQ4D31006 or compatible part.

The x-direction from centerline to centerline of connectors in the same U is 28.5mm. The y-direction from centerline to centerline of connectors in the same 2U space of the cage is 42.05mm. The vertical spacing between connectors in separate but adjacent 2U spaces is 46.05mm (to account for the sheet metal thickness in the cage).

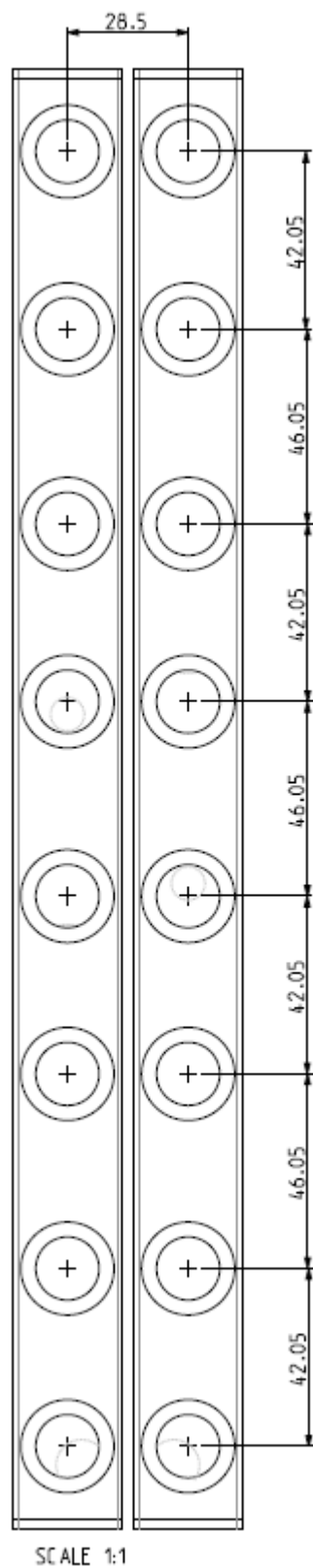


Figure 20. Manifold connector dimensions (from front)

3.3.4. Two-phase manifold

Two-phase manifold specified dimensions

The x-direction from centerline to centerline of connectors in the same U is 27.0mm. The y-direction from centerline to centerline of connectors in the same 2U space of the cage is 42.05mm. The vertical spacing between connectors in separate but adjacent 2U spaces is 46.05mm (to account for the sheet metal thickness in the cage).

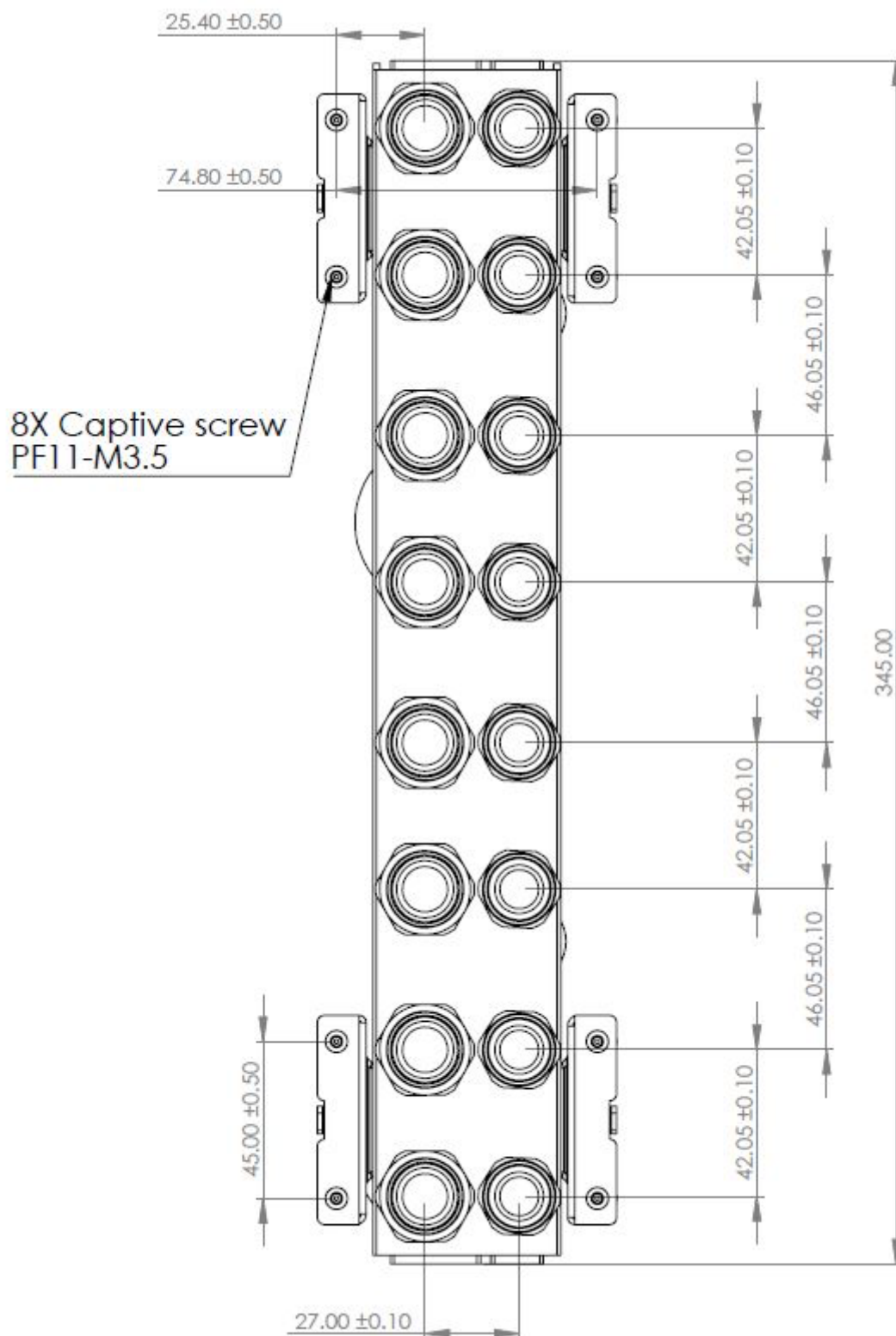


Figure 21. Manifold and connectors location dimensions (from front)

3.4. Facility water temperature

For both single-phase and two-phase liquid cooling, the expected facility water temperature shall be ASHRAE W27 or lower (eg: W17). The implied temperature to Open19 V2 bricks is 27C plus a CDU approach (expected temperature is 30-33C to a single-phase cold plate.)

These temperature expectations are an alignment with the OCP Coolant Temperatures for durable data center design.

Chapter 4. Open19 V2 Cable System Specification

4.1. General

The Open19 platform simplifies installation with blind-mate data and power interfaces. To facilitate inter-operability, aspects of the power shelf, brick, and data cable assemblies shall be defined. The cable systems are all completely passive. Components inheriting existing specifications (eg: SFP, QSFP, QSFP-DD) shall be compliant with the industry definitions.

4.2. Open19 Power Cabling

The Open19 V2 power distribution cabling groups each half-cage (1/2 wide 1U) into a power connector on the power shelf. Each assembly is intended for a single cage. The V2 specification relaxes the requirement on the location of the power shelf and moves the assembly definition to an implementation guide. With the shift to +48V DC power, the signal integrity on the data cable at 50G-PAM4 typically provides implementation constraints. Power cable assemblies are expected to be longer with more variations than the V1 specification.

Each cabling system is built for 8-channel aggregation. Each bundle is an array of power distribution cables (supply and return) and carries +48V DC (+51V nominal) to the bricks. Power shelf connectors are rated to carry 70A. Brick connectors may carry 35A or 70A. Cables shall be sized appropriately to handle the current and length.

4.2.1. Open19 Power shelf connector

Power shelves may implement any power connector that meets brick power requirements and brick inter-operability. The Open19 community has developed a power shelf connector set utilizing the Molex Coeur Socket (tm). Any power harness mating to the Couers Socket and Ten60 blade connector shall use the following design. Power shelf designs that do NOT implement the Coeur Socket, or interoperable connector, may propose alternate designs for inclusion to the specification. Brick connectors shall always interop with the 70A Ten60 blade connector.

Power shelf connector

The power shelf connector is a 70A rated assembly. It shall be used for all implementations 70A and less per power feed to bricks. A detect pin is used to indicate the type of assembly connected to the power shelf.

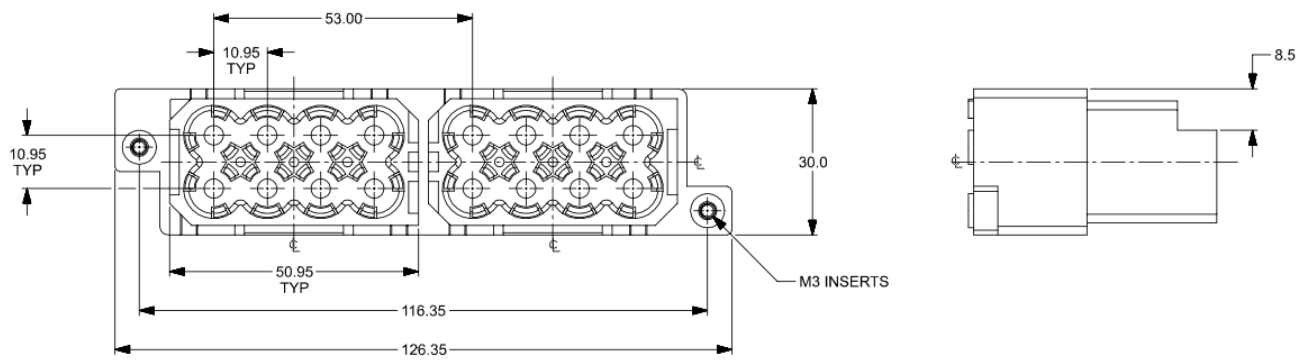


Figure 22. Power shelf connector

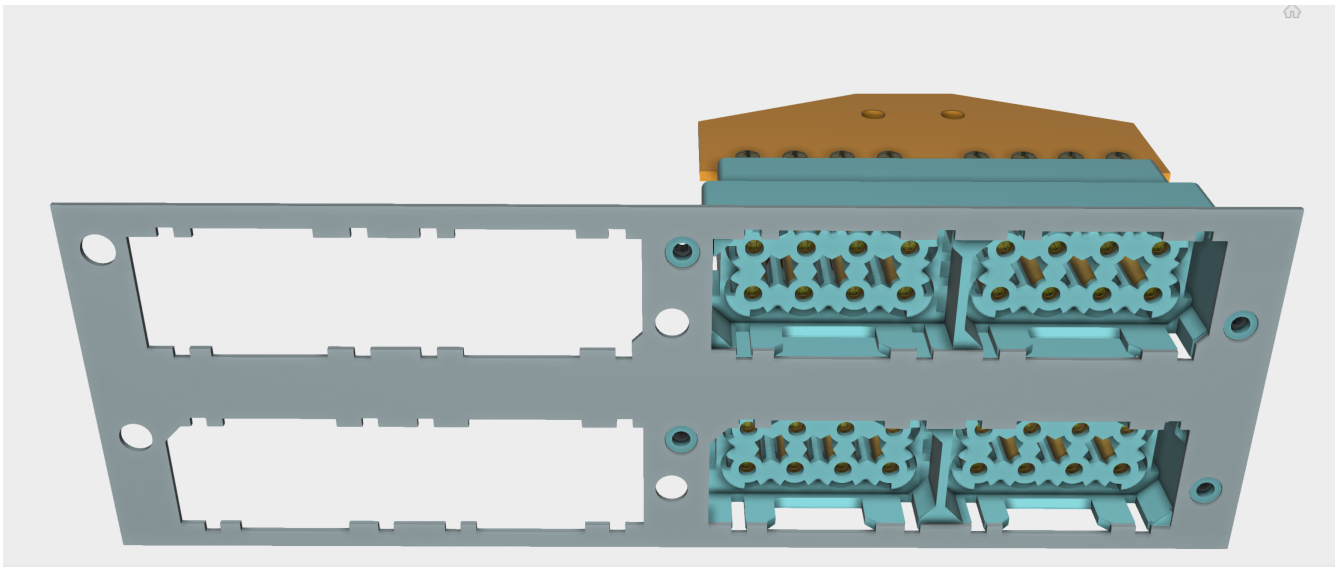


Figure 23. Power shelf connector mounted in panel, upper view

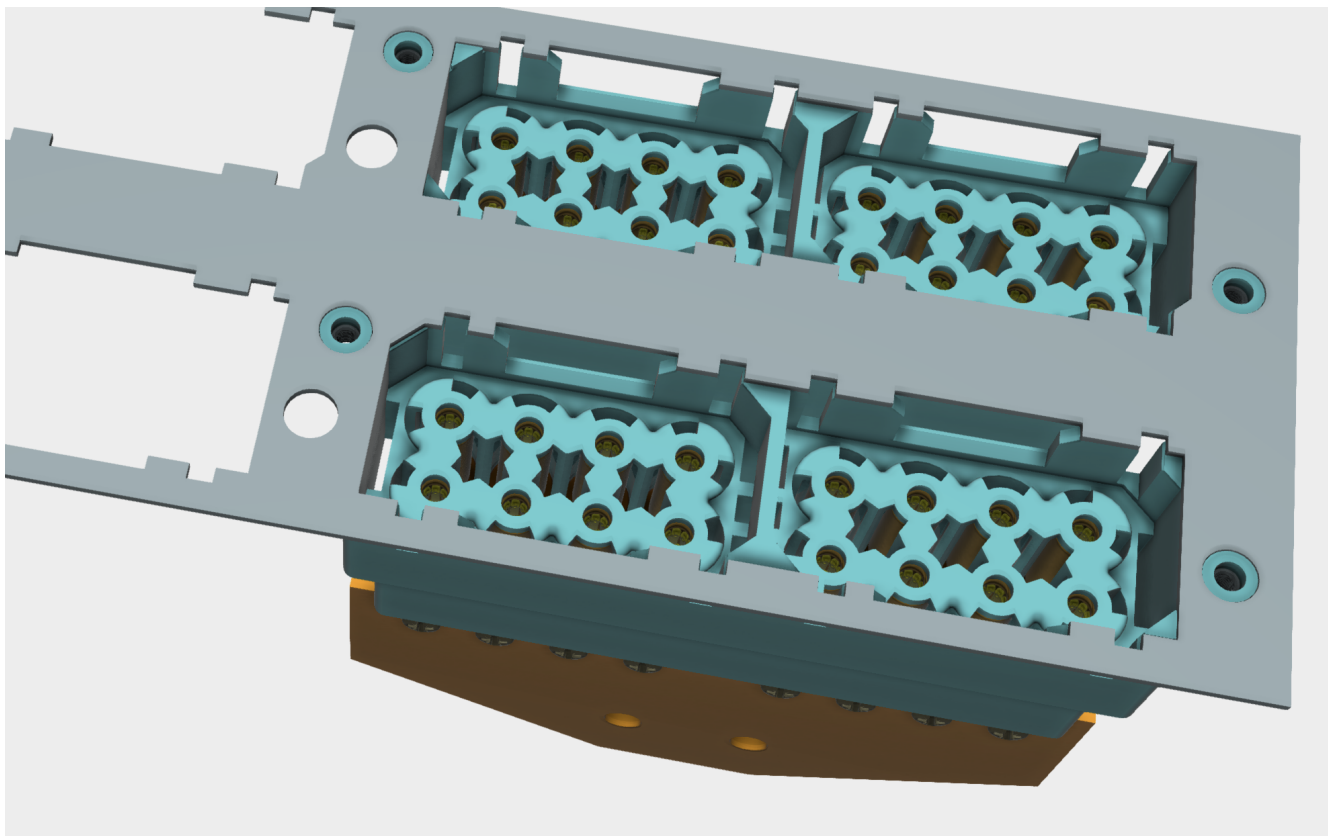


Figure 24. Power shelf connector mounted in panel, lower view

Power shelf universal input connectors

The power shelf shall use input connectors common to OCP Open Rack V3. Refer to the [Open Rack V3 Shelf AC Input connector V1.0](#) for detailed information.

4.3. Open19 data cable assembly

Open19 V2 relaxes the data cable requirements and has moved the cables into implementation specific designs. The brick data connector provides definitions for all pins and usage, in order to enable interoperability for all brick designs.

The data cable shall provide, as needed, the EEPROM required for SFP, QSFP, and QSFP-DD connectors.

4.3.1. SFP breakout implementation

The SFP28 breakout implementation supports four (4) SFP28 connections from each brick connector.

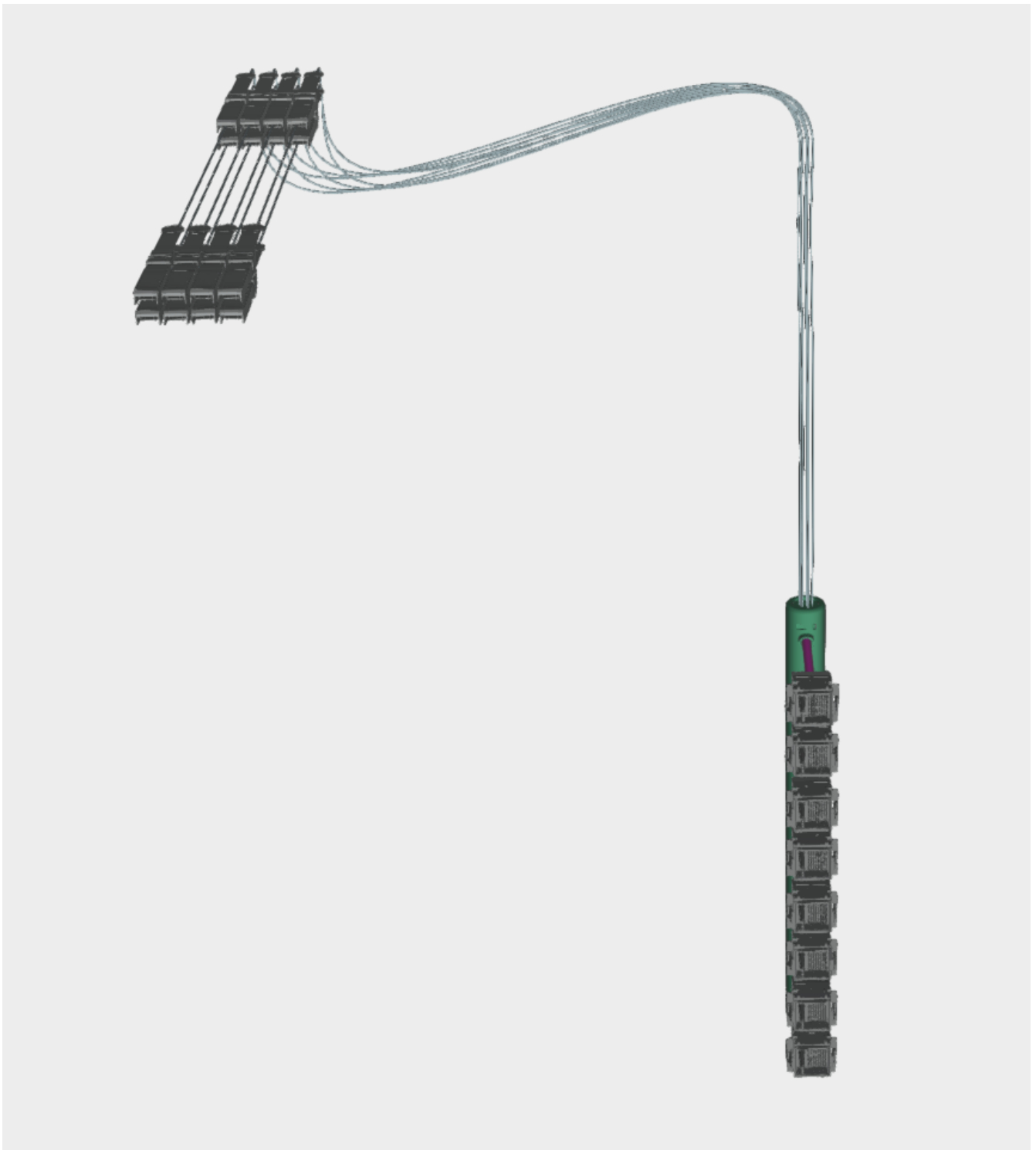


Figure 25. SFP28 breakout cable

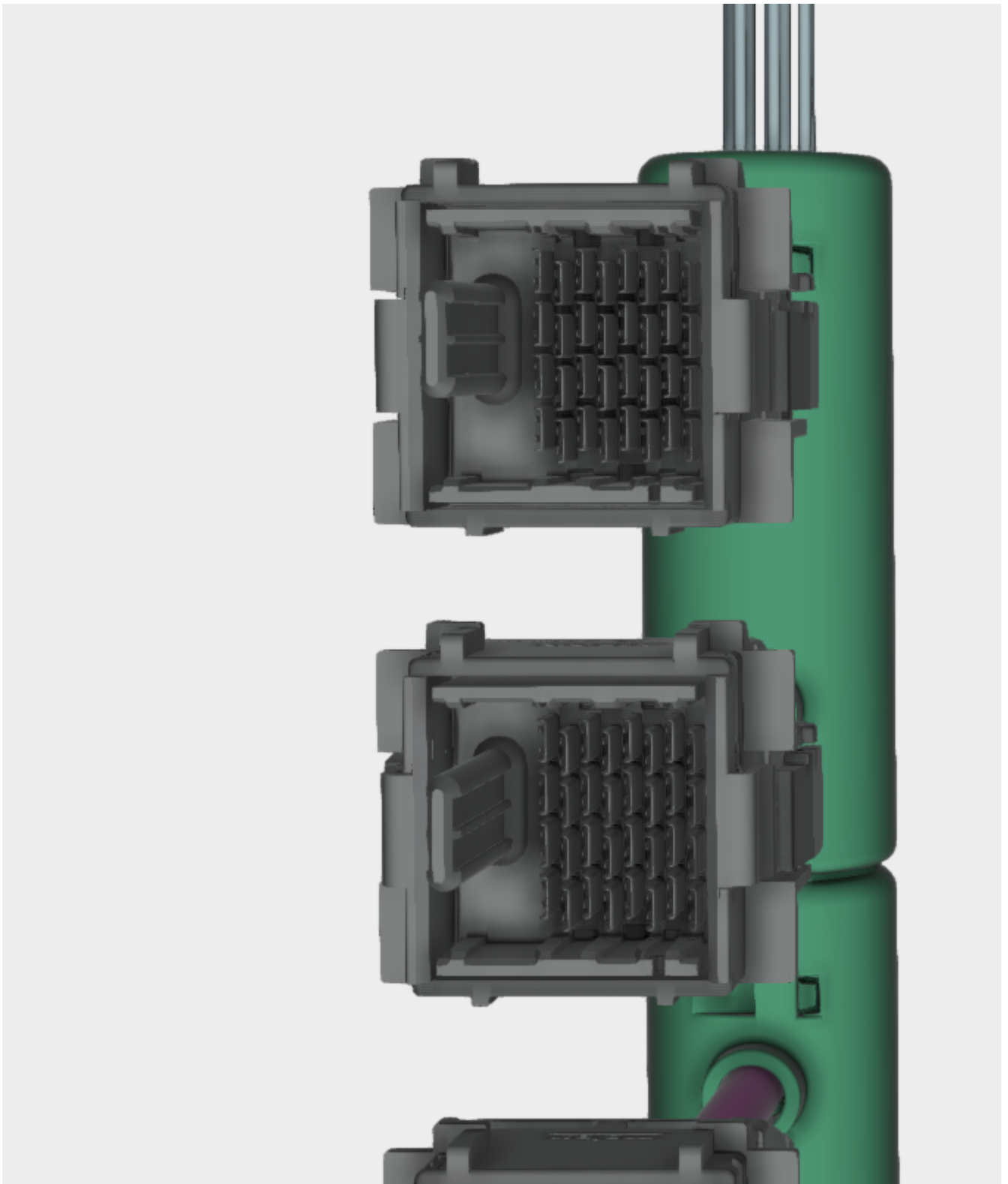


Figure 26. Cage Impel attachment

Chapter 5. Environmental Requirements

The Open19 platform can operate in any data center environment with no special restrictions nor requirement. The two environmental requirements are:

- Ambient air temperature of 10-40°C at sea-level to 1000m elevation
- Unobstructed front and back flow

Noted that while an enclosed rack is also supported for the Open19 platform, the required airflow per disaggregated component in the platform needs to be reviewed separately in case of obstructed airflow.